Yuzhen Song

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EDUCATION

B.Eng. in Robotics Engineering Aug 2023 – Present
Southern University of Science and Technology Shenzhen, China

- **Overall GPA: 3.85**/4.0 , Weighted Avg: **92.63**/100
- **Core Courses:** Robot Operating System(100), Mechanics of Materials M(100), Robot Modeling and Control(97), Mechanical Design(97), CAD and Engineering Drawing(93), Intelligent Manufacturing System Technology(95), Signals and Systems(95), Theoretical Mechanics I-B(93), Cognitive Practice in Manufacturing Engineering(93), Fundamentals of Control Engineering(90)
- **Leadership:**
  - President, Student Organizations Council (SOC)
  - President, Robotics Club
  - Class President, Dept. of Mechanical & Energy Engineering

RESEARCH EXPERIENCES

Southern University of Science and Technology **Shenzhen, China**
Undergraduate Researcher, Advisor: Prof. Boyu Zhou *10/2025 – 12/2025*

- **Project: Autonomous UAV Mission Navigation via Fast-Planner Framework**
  - Deployed the **Fast-Planner** framework on **ROS** to generate real-time, kinodynamically feasible trajectories within randomized point-cloud and cuboid obstacle environments.
  - Engineered a **custom circular benchmark track** in **RViz** featuring ring-shaped point clouds; developed **autonomous mission logic** to execute sequential waypoint traversal and return-to-origin navigation.
  - Evaluated system performance through consistent autonomous traversal of constrained circular circuits, validating the efficacy of the planner's obstacle avoidance and trajectory smoothing in user-defined scenarios.

Undergraduate Researcher, Advisor: Prof. Chenglong Fu *12/2025 – Present*

- **Project: Slope Center-of-Mass (CoM) Assistance System for Human Gait**
  - Designed an experimental testbed on an aluminum profile frame, integrating a motor-driven winch and a **compound pulley system** (fixed & movable) to deliver **dynamic, regulated traction (up to 200 N peak)** with **<5% force control error** to the human Center of Mass (CoM).
  - Developed an **STM32-based embedded control architecture** for high-fidelity motor torque and speed regulation, synchronized with force sensor telemetry to ensure accurate assistance delivery during slope walking.
  - Performed **biomechanical model validation** by analyzing the correlation between slope gradients and gait parameters (e.g., **step length**, impulses, and CoM fluctuations) across multi-slope kinematic datasets.
  - Evaluated the relationship between metabolic energy reduction and assistance strategies, confirming the model's efficacy in predicting energy-efficient walking patterns and achieving an estimated **20–30% reduction** in metabolic cost under various incline conditions.
  - *Manuscript in preparation; target submission to **IROS**.*

Undergraduate Researcher, Advisor: Prof. Zhao Tang *07/2025 – 11/2025*

- **Project: Sim-to-Real Infrastructure for Closed-Loop Dexterous Hand**
  - Refined the **URDF** model of a multi-fingered robotic palm; optimized closed-loop kinematic constraints and collision geometries to ensure physical accuracy within high-fidelity physics engines.
  - Established robust simulation environments in **Webots** and **NVIDIA Isaac Sim**, constructing a foundational pipeline designed for future **Reinforcement Learning (RL)** grasping policy training.
  - Configured contact dynamics and joint parameters within simulation, laying the essential technical framework for future **Sim-to-Real** policy transfer and closed-loop feedback experiments.

PROJECTS

Project Leader, Advisor: Prof. Wende Ke *09/2025 – 12/2025*

- **Project: ROS2-Based Mobile Manipulator with Multimodal HRI**
  - Architected a two-wheeled differential drive mobile base integrated with a 4-DOF manipulator in SolidWorks; optimized its **URDF/xacro** model with Gazebo plugins for LiDAR, **RGB camera**, and IMU.
  - Deployed **ROS2 Nav2** and LiDAR-based **SLAM** for autonomous mapping and navigation; engineered an **Image-Based Visual Servoing** pipeline combining HSV color segmentation and dual-loop **PID control**, achieving $\pm$**5-pixel** ($\approx\pm$5 mm) horizontal alignment accuracy for robust dynamic grasping.
  - Integrated an **Extended Kalman Filter (EKF)** (*robot_localization*) to fuse IMU and wheel odometry data, effectively mitigating sensor drift and improving localization stability.
  - Implemented a multimodal Human-Robot Interaction (HRI) interface using **MediaPipe** for 21-keypoint gesture recognition and **Vosk API** for offline voice-command execution.
  - *GitHub Repository: https://github.com/amysongswimkim/ros2-mobile-manipulator*

Project Leader, Advisor: Prof. Dong Lu, Yuanjian Huang *11/2024 – 08/2025*

- **Project: Autonomous Intelligent Rescue Vehicle (National Competition)**
  - Programmed **STM32 (C/C++)** to process visual telemetry via UART and orchestrated a **Finite State Machine (FSM)** for chassis kinematics and PWM-based servo grasping; developed **active perception logic** (dynamic camera pitching) to verify target capture and optimize manipulation precision.
  - Curated a domain-specific dataset utilizing **X-AnyLabeling** for automated annotation; trained and deployed a custom **YOLOv11** object detection model, achieving robust real-time multi-class recognition (targets, safe zones) under variable lighting.
  - **Achievement: Led the team** to win the **National Second Prize** and **Provincial First Prize (3rd out of 46 teams)** in the China Undergraduate Engineering Practice and Innovation Ability Competition.

Project Leader, Advisor: Prof. Chenglong Fu *03/2024 – 06/2024*

- **Project: CV-Guided 6-DOF Robotic Sketching System**
  - Engineered a custom **compliant end-effector** with a spring-buffered linear guide, enabling adaptive Z-axis pressure regulation to replicate authentic brushstrokes and leverage the manipulator's spatial dexterity.
  - Developed an **OpenCV** vision pipeline (skeletonization, connected-component analysis) and a greedy Euclidean search algorithm to extract sub-pixel contours and generate optimized continuous toolpaths.
  - Solved 6-DOF **inverse kinematics (IK)** via **D-H parameters** to orchestrate high-precision physical sketching on planar paper via **MATLAB**; expanded the framework to generate and validate 3D spatial trajectories for curved surfaces (e.g., cylinders) within **Simulink**.

SKILLS

- **Programming Languages:** Proficient in C Language, C++, Python and MATLAB.
- **Mechanical Design and Manufacturing Software:** AutoCAD, SolidWorks, Edgecam, Abaqus
- **Integrated Development Environments (IDEs):** Visual Studio, VScode, Keil5, Arduino
- **Robot Simulation and Visualization:** Gazebo, Webots, Issac Sim, Rviz2

HONORS & AWARDS

- National Second Prize, China Undergraduate Engineering Practice and Innovation Ability Competition ***2025***
- Provincial First Prize, Guangdong Undergraduate Engineering Practice and Innovation Ability Competition ***2025***
- First-Class Scholarship for Outstanding Students (2024-2025 Academic Year) ***2025***
- Provincial Second Prize, Guangdong Undergraduate Mechanical Innovational Competition ***2025***
- Outstanding Student Award ***2024, 2025***
- Second-Class Scholarship for Outstanding Students (2023-2024 Academic Year) ***2024***